CRICKSTATS

CRICKET PERFORMANCE INTELLIGENCE PLATFORM

COLLEGE: JAWAHAR SCIENCE COLLEGE, NEYVELI

DEPARTMENT: DEPARTMENT OF COMPUTER SCIENCE AND APPLICATION

COURSE: BACHELOR OF COMPUTER APPLICATION

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ABSTRACT

ABSTRACT

CRICKSTATS is a Salesforce-based Cricket Performance Intelligence Platform designed to manage, store, and analyze cricket player statistics across multiple matches. The system is developed using Salesforce Lightning Experience and leverages declarative tools such as custom objects, master-detail relationships, roll-up summary fields, formula fields, validation rules, reports, dashboards, and flow automation.

The platform automates key performance metrics including batting strike rate, bowling economy rate, player averages, and overall performance ranking. It also ensures data accuracy through validation rules and enhances user experience through a custom Lightning App interface. This project demonstrates the effective use of Salesforce data modeling, automation, and analytics tools in building a real-world sports performance management system.

INTRODUCTION

INTRODUCTION

Cricket analytics play a vital role in evaluating player performance and assisting teams in strategic decision-making. Tracking statistics such as runs scored, wickets taken, strike rate, economy rate, and match averages help coaches and analysts identify player strengths and areas for improvement.

Traditional manual record-keeping methods are often inefficient and prone to human errors. With increasing demand for data-driven decision-making, there is a need for an automated system that can store structured player data and generate real-time analytical insights.

The CRICKSTATS system was developed to address this requirement by using Salesforce's declarative tools to build a centralized platform for managing cricket performance data. The system automatically calculates key metrics, ensures data accuracy, and presents insights through interactive dashboards and reports.

The primary objectives of the system are:

- To store and manage player performance data
- To automate calculation of cricket statistics
- To ensure data integrity through validation rules
- To visualize performance insights using reports and dashboards
- To provide an interactive user interface through flow automation

PROBLEM STATEMENT

PROBLEM STATEMENT

In cricket performance analysis, maintaining accurate player statistics manually can be time-consuming and error prone. Coaches and analysts need reliable data to evaluate player performance across multiple matches. However, manual record-keeping methods often lead to inconsistencies and difficulty in calculating complex performance metrics such as strike rate, bowling economy rate, and overall performance rankings.

Additionally, without a centralized system, retrieving historical performance data and comparing players becomes inefficient. Therefore, there is a need for an automated and structured system that can store match data, calculate performance metrics automatically, and provide analytical insights through visual reports and dashboards.

The CRICKSTATS platform addresses this problem by providing an automated Salesforce-based system that efficiently manages cricket performance data and delivers meaningful analytics.

FEASIBILITY STUDY

FEASIBILITY STUDY

A feasibility study evaluates whether the proposed system is practical and achievable.

TECHNICAL FEASIBILITY:

The project is technically feasible because it uses Salesforce Developer Edition, which provides built-in tools for data modeling, automation, reporting, and analytics. The platform supports declarative development, allowing complex business logic to be implemented without programming.

OPERATIONAL FEASIBILITY:

The system is user-friendly and operates within the Salesforce Lightning interface. Users can easily create player records, enter match statistics, and view performance analytics without requiring technical expertise.

ECONOMIC FEASIBILITY:

The project is economically feasible because Salesforce Developer Edition is available free of cost for development and educational purposes. Therefore, the system can be implemented without additional infrastructure expenses.

SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE

The CRICKSTATS system consists of two main custom objects:

1. Player (Parent Object)
   - Stores personal details and aggregated performance statistics.
2. Match Performance (Child Object)
   - Stores match-by-match batting, bowling, and fielding statistics.

A Master-Detail Relationship connects Match Performance to Player. This relationship enables roll-up summary fields that automatically calculate player performance metrics.

SETUP > OBJECT MANAGER

Player

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters

Search Layouts

- List View Button Layout
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules
- Conditional Field Formatting

Fields & Relationships

22 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE
AVG Score	AVG_Score__c	Formula (Number)
Batting Style	Batting_Style__c	Picklist
Bowling Style	Bowling_Style__c	Picklist
Created By	CreatedById	Lookup(User)
Date of Birth	Date_of_Birth__c	Date
Last Modified By	LastModifiedById	Lookup(User)
Matches Played	Matches_Played__c	Roll-Up Summary (COUNT Match Performance)
Nationality	Nationality__c	Picklist
Owner	OwnerId	Lookup(User,Group)
Performance Score	Performance_Score__c	Formula (Number)
Player Name	Name	Text(80)
Player Number	Player_Number__c	Text(3)
Profile Picture	Profile_Picture__c	Rich Text Area(32768)
Role	Role__c	Picklist
Team	Team__c	Picklist
Total 4's	Total_4_s__c	Roll-Up Summary (SUM Match Performance)
Total 6's	Total_6_s__c	Roll-Up Summary (SUM Match Performance)
Total Catches	Total_Catches__c	Roll-Up Summary (SUM Match Performance)
Total Overs	Total_Overs__c	Roll-Up Summary (SUM Match Performance)
Total Runs	Total_Runs__c	Roll-Up Summary (SUM Match Performance)
Total Stumpings	Total_Stumpings__c	Roll-Up Summary (SUM Match Performance)
Total Wickets	Total_Wickets__c	Roll-Up Summary (SUM Match Performance)

SETUP > OBJECT MANAGER

Match Performance

- Details
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- Scoping Rules
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- Flow Triggers
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- Conditional Field Formatting

Fields & Relationships

10 items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE
Balls Faced	Balls_Faced__c	Number(5, 0)
Catches	Catches__c	Number(3, 0)
Created By	CreatedById	Lookup(User)
Economy Rate	Economy_Rate__c	Formula (Number)
Fours	Fours__c	Number(3, 0)
Last Modified By	LastModifiedById	Lookup(User)
Maiden Overs	Maiden_Overs__c	Number(3, 0)
Match Performance Name	Name	Text(80)
Match Result	Match_Result__c	Picklist
Overs Bowled	Overs_Bowled__c	Number(3, 1)
Player	Player__c	Master-Detail(Player)
Record Type	RecordTypeId	Record Type
Runs Conceded	Runs_Conceded__c	Number(5, 0)
Runs Scored	Runs_Scored__c	Number(5, 0)
Sixes	Sixes__c	Number(3, 0)
Strike Rate	Strike_Rate__c	Formula (Number)
Stumpings	Stumpings__c	Number(3, 0)
Wickets Taken	Wickets_Taken__c	Number(3, 0)

SETUP > OBJECT MANAGER

Match Performance

- Details
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- Buttons, Links, and Actions
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- Restriction Rules
- Scoping Rules
- Object Access
- Triggers

Match Performance Custom Field

Player

Back to Match Performance

Validation Rules (0)

Custom Field Definition Detail — Edit | Set Field-Level Security | View Field Accessibility | Where is this used?

Field Information

Field Label	Player	Object Name	Match Performance
Field Name	Player	Data Type	Master-Detail
API Name	Player__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Adithyan Team, 3/28/2026, 12:53 AM	Modified By	Adithyan Team, 3/28/2026, 12:53 AM

Master-Detail Options

Related To	Player	Child Relationship Name	Match_Performances
Related List Label	Match Performances		
Sharing Setting	Read/Write: Allows users with at least Read/Write access to the Master record to create, edit, or delete related Detail records.		
Reparentable Master Detail			

Lookup Filter

No lookup filters defined.

Validation Rules — New

No validation rules defined.

DATA MODEL DESIGN

PLAYER OBJECT

The Player object stores personal information and aggregated statistics.

Key fields include:

- Player Name
- Player Number
- Date of Birth
- Role
- Batting Style
- Bowling Style
- Nationality
- Team
- Profile Picture

Roll-up summary fields aggregate match statistics.

Examples:

- Matches Played
- Total Runs
- Total Wickets
- Total 4's
- Total 6's
- Total Overs
- Total Catches
- Total Stumpings

SETUP > OBJECT MANAGER

Player

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
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Fields & Relationships

22 Items, Sorted by Field Label

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Total Overs	Total_Overs_c	Roll-Up Summary (SUM Match Performance)
Total Runs	Total_Runs_c	Roll-Up Summary (SUM Match Performance)
Total Stumpings	Total_Stumpings_c	Roll-Up Summary (SUM Match Performance)
Total Wickets	Total_Wickets_c	Roll-Up Summary (SUM Match Performance)

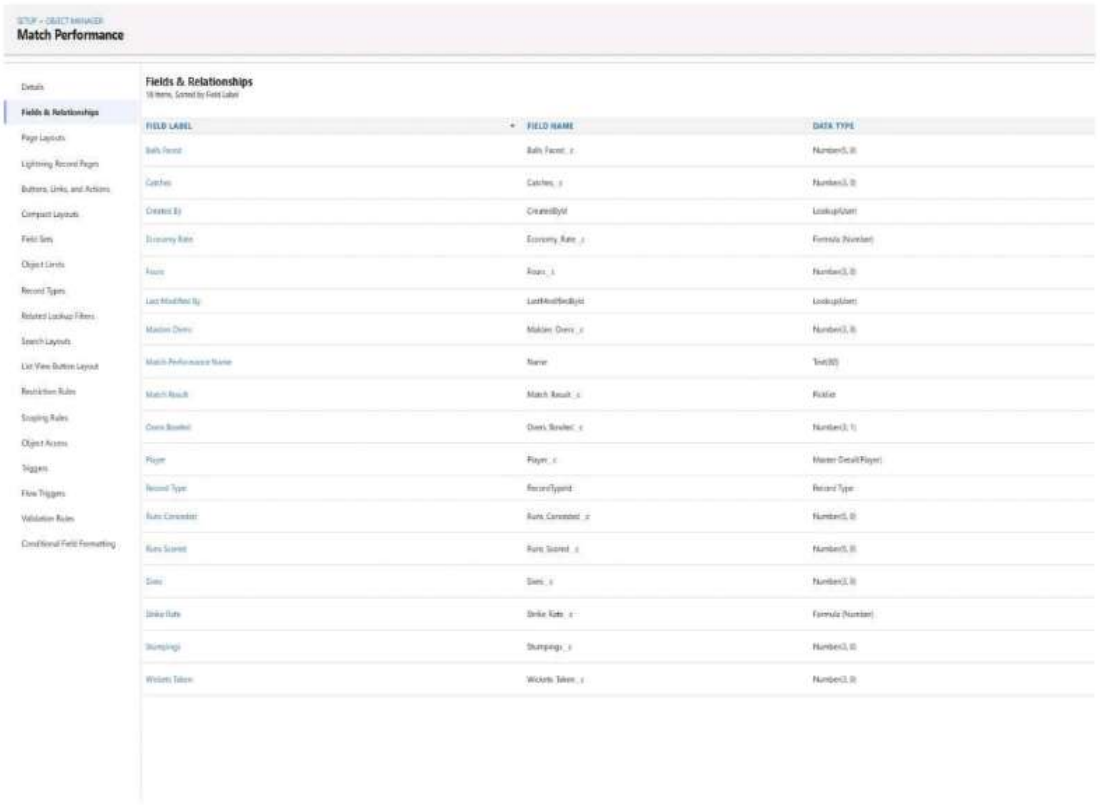
MATCH PERFORMANCE OBJECT

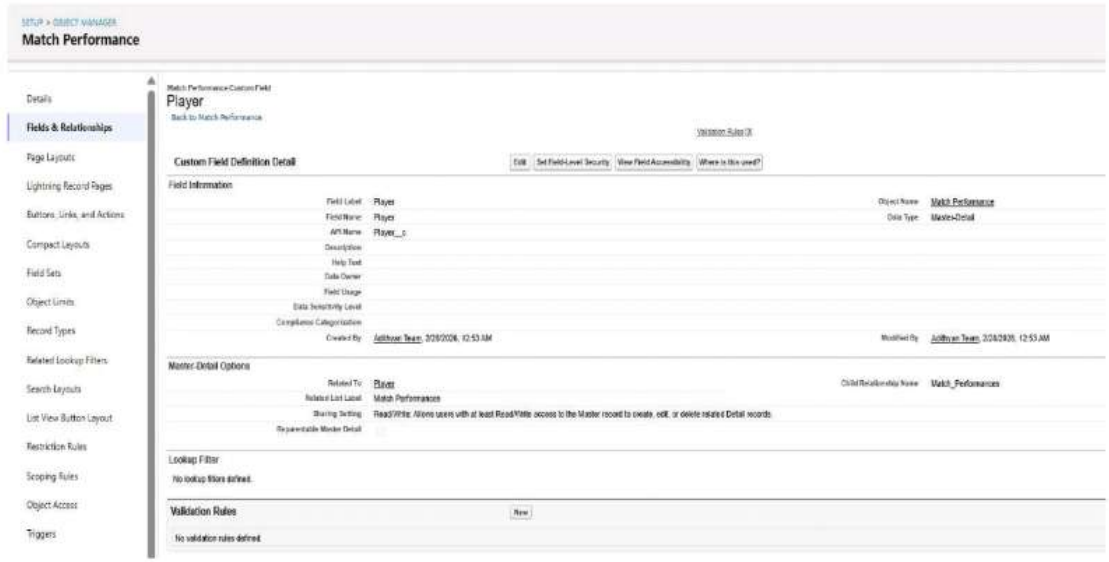
This object records match-specific statistics.

Key fields include:

- Runs Scored
- Balls Faced
- Fours
- Sixes
- Overs Bowled
- Wickets Taken
- Runs Conceded
- Maiden Overs
- Catches
- Stumpings

Formula fields calculate advanced metrics such as Strike Rate and Economy Rate.


SETUP > OBJECT MANAGER
Match Performance
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Validation Rules
Conditional Field Formatting
Fields & Relationships
18 Items, Sorted by Field Label
FIELD LABEL
FIELD NAME
DATA TYPE


SETUP > OBJECT MANAGER
Match Performance
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Triggers
Match Performance Custom Field
Player
Back to Match Performance
Custom Field Definition Detail
Edit
Set Field-Level Security
View Field Accessibility
Where is this used?
Field Information
Field Label Player
Field Name Player
API Name Player__c
Description
Help Text
Data Owner
Field Usage
Data Sensitivity Level
Compliance Categorization
Created By Adithyan Team, 3/28/2026, 12:53 AM
Object Name Match Performance
Data Type Master-Detail
Modified By Adithyan Team, 3/28/2026, 12:53 AM
Master-Detail Options
Related To Player
Related List Label Match Performances
Sharing Setting Read/Write. Allows users with at least Read/Write access to the Master record to create, edit, or delete related Detail records.
Reparentable Master Detail
Child Relationship Name Match_Performances
Lookup Filter
No lookup filters defined.
Validation Rules
New
No validation rules defined.

BUSINESS LOGIC IMPLEMENTATION

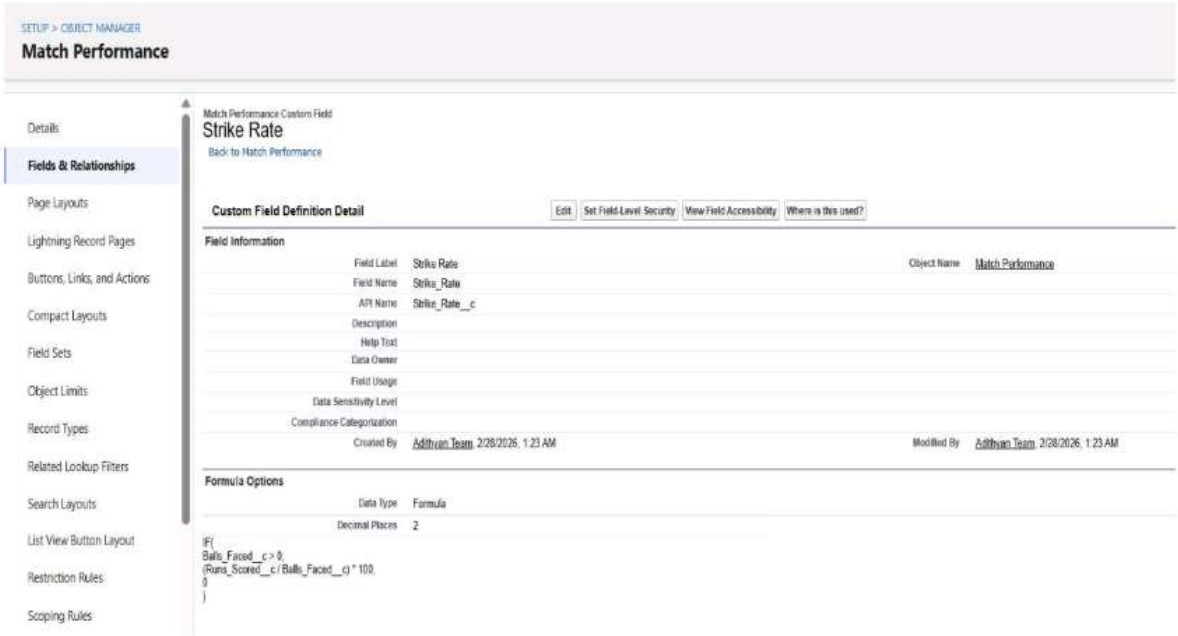
MATCH PERFORMANCE OBJECT

The system uses formula fields and roll-up summary fields to automate cricket performance calculations.

STRIKE RATE FORMULA:

```
IF(
Balls_Faced__c > 0,
(Runs_Scored__c / Balls_Faced__c) * 100,
0
)
```

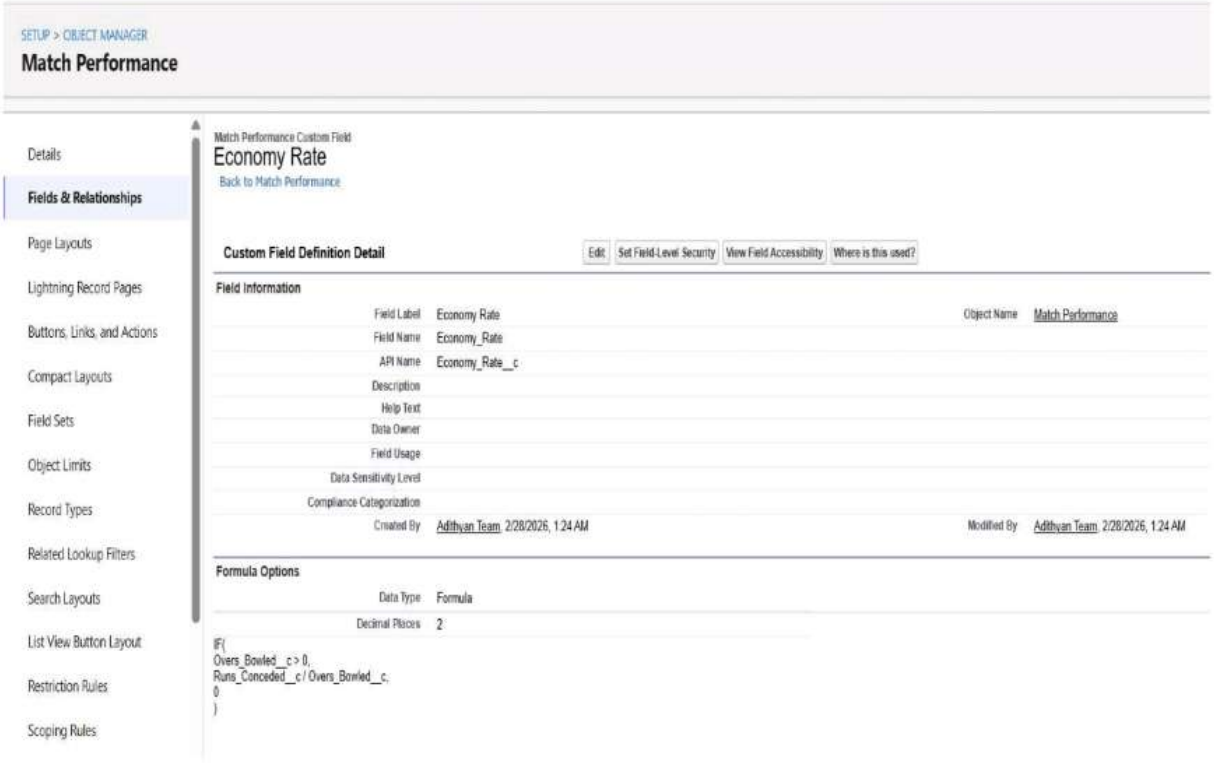
This formula calculates the batting strike rate while preventing division-by-zero errors.



ECONOMY RATE FORMULA:

```
IF(
Overs_Bowled__c > 0,
Runs_Conceded__c / Overs_Bowled__c,
0
)
```

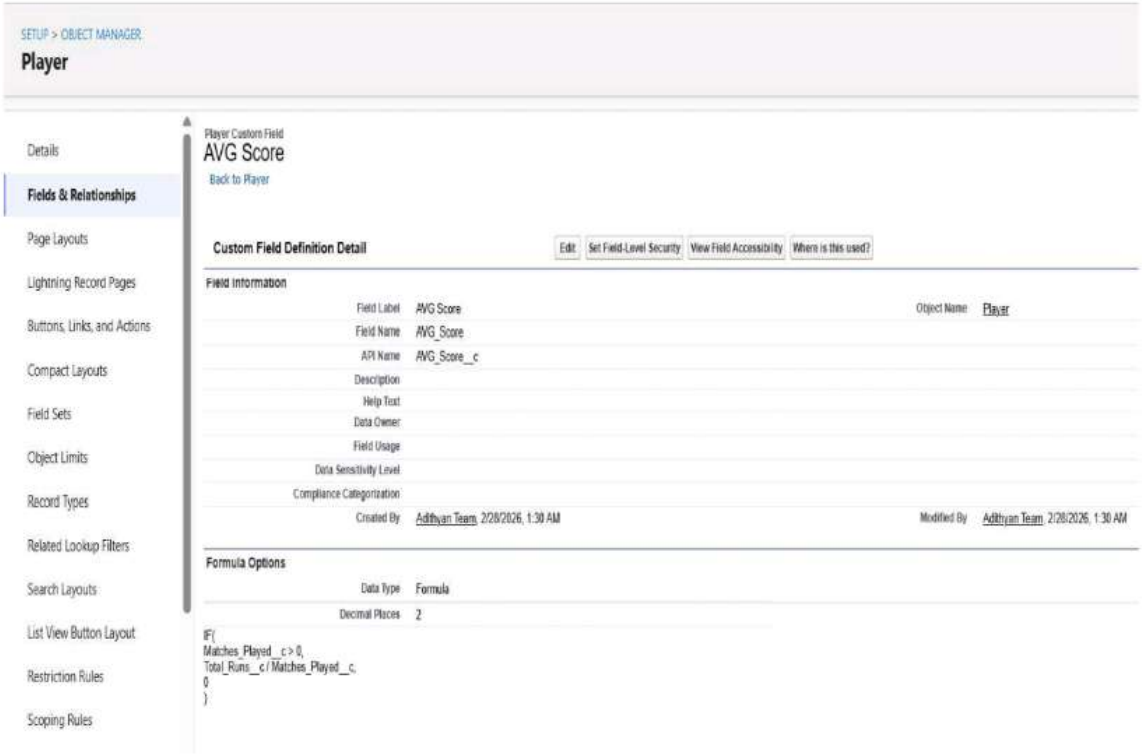
This formula calculates the bowling economy rate.



AVERAGE SCORE FORMULA:

```
IF(
Matches_Played__c > 0,
Total_Runs__c / Matches_Played__c,
0
)
```

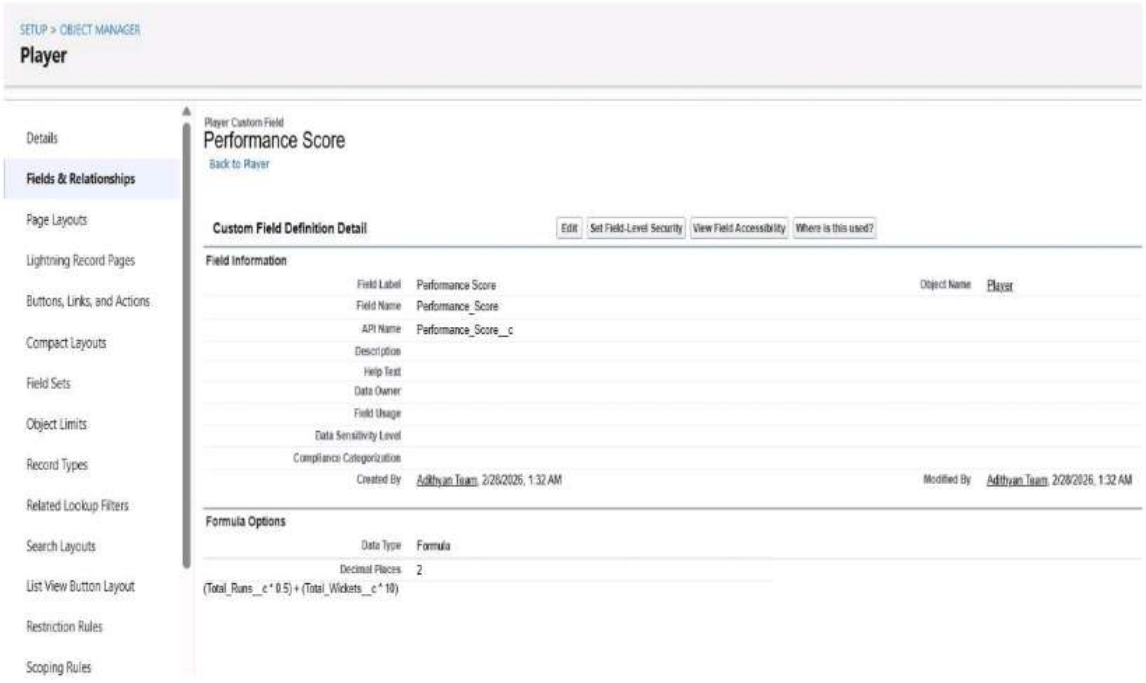
This formula calculates the average runs scored per match.



PERFORMANCE SCORE FORMULA:

(Total_Runs__c * 0.5) + (Total_Wickets__c * 10)

This formula generates a weighted performance ranking based on batting and bowling contributions.



DATA VALIDATION RULES

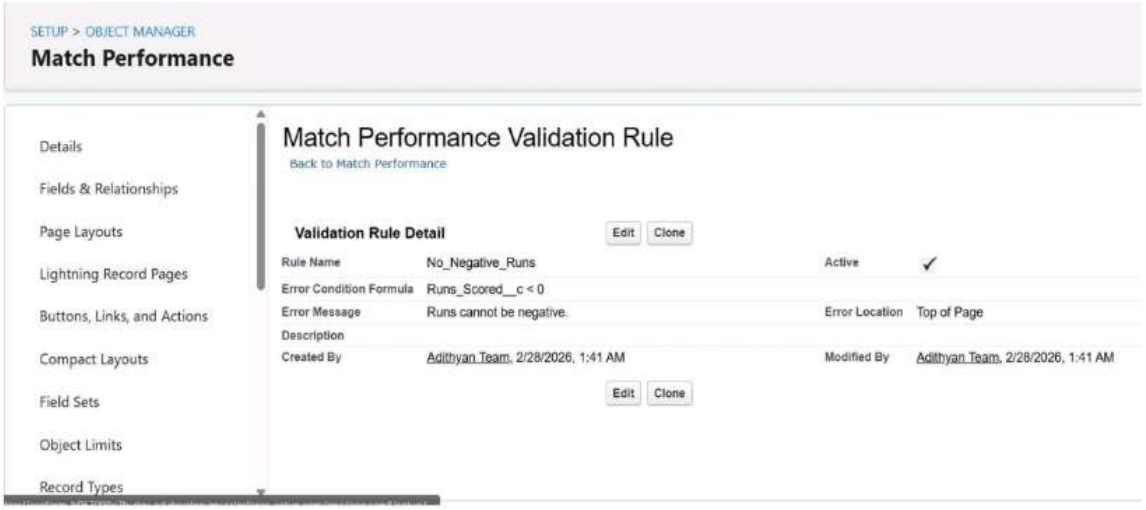
DATA VALIDATION RULES

Validation rules ensure that unrealistic or invalid data cannot be entered.

NEGATIVE RUNS RESTRICTION:

Runs_Scored__c < 0

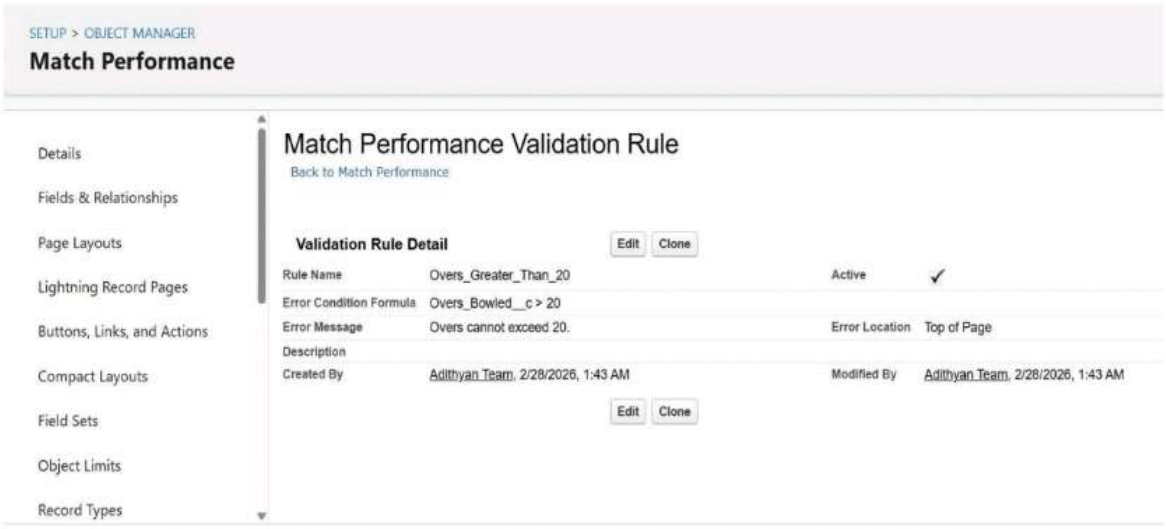
Error Message: Runs cannot be negative.



OVERS LIMIT RULE:

Overs_Bowled__c > 20

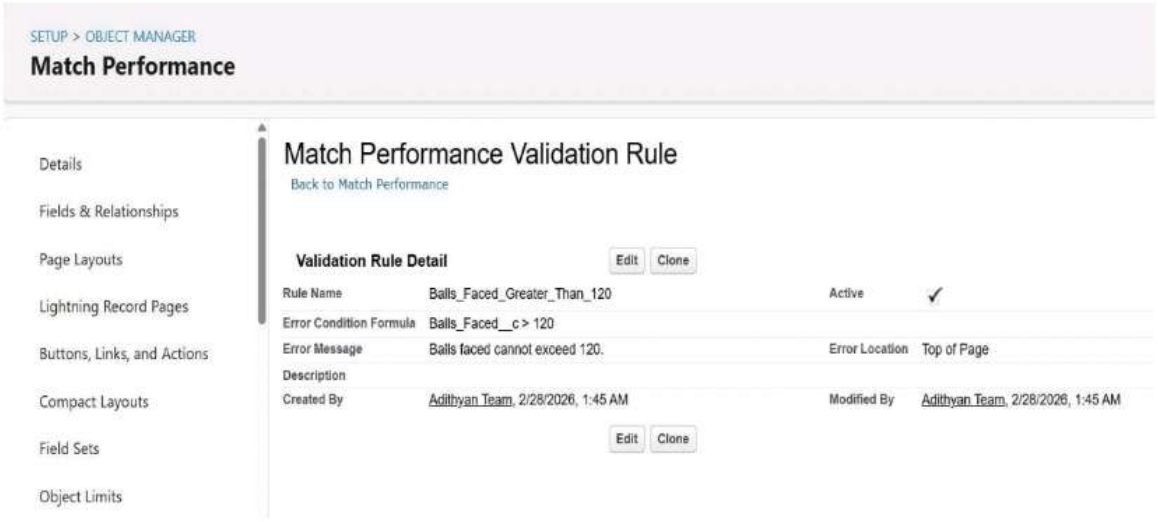
Error Message: Overs cannot exceed 20.



BALLS FACED LIMIT RULE:

Balls_Faced__c > 120

Error Message: Balls Faced cannot exceed 120.



RECORD TYPES IMPLEMENTATION

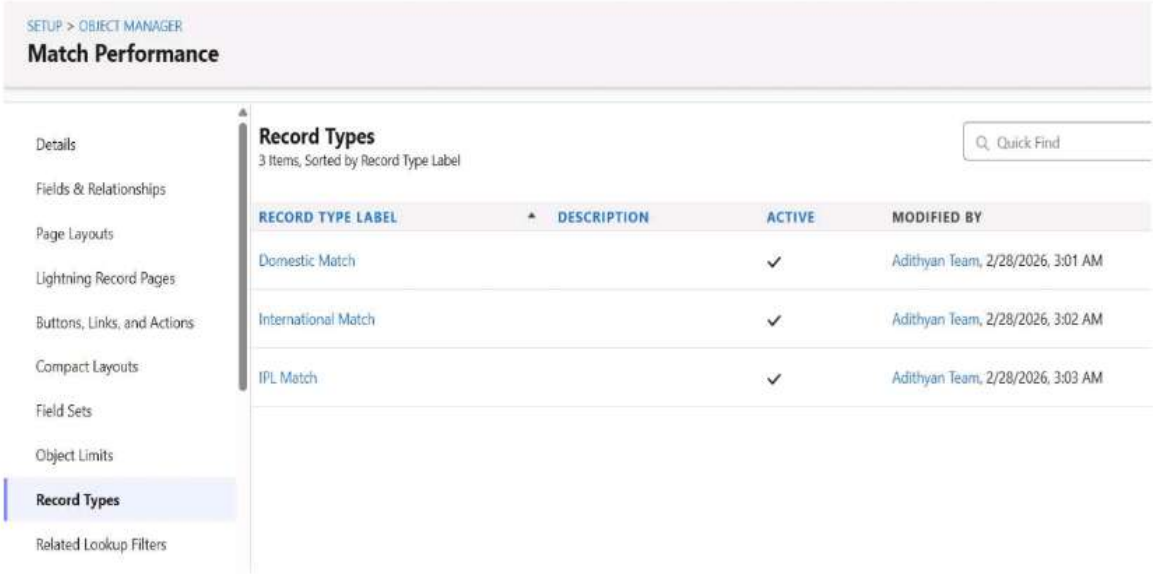
RECORD TYPES IMPLEMENTATION

Record types categorize match performances.

Three record types were created:

- Domestic Match
- International Match
- IPL Match

This allows users to classify match data based on competition format.



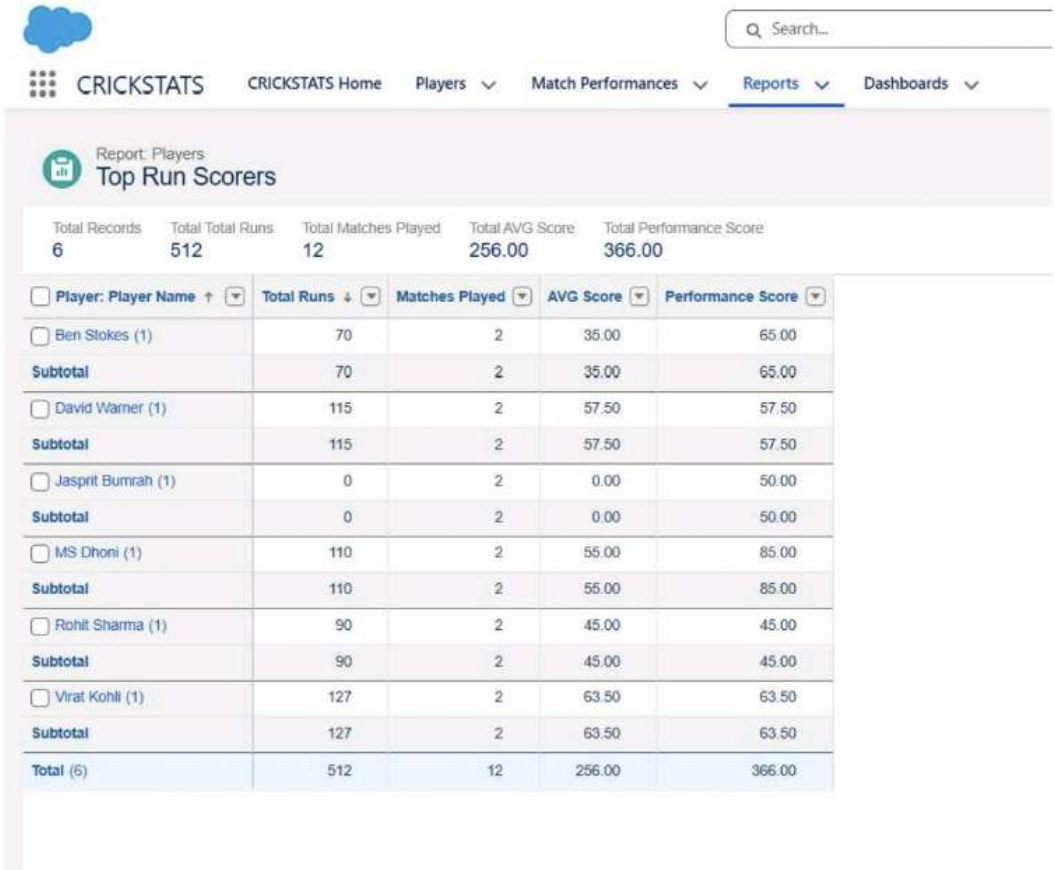
REPORTS

REPORTS

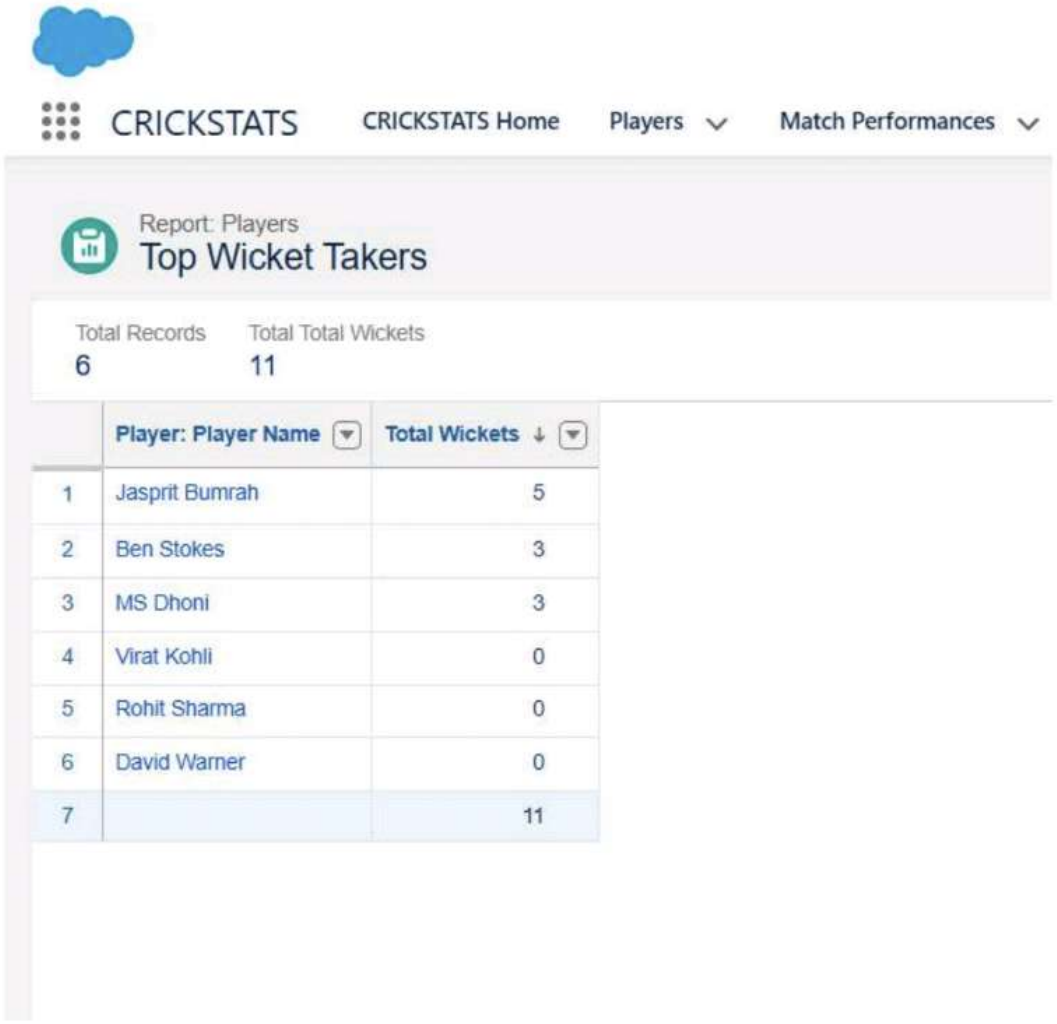
Several reports were created to analyse player statistics.

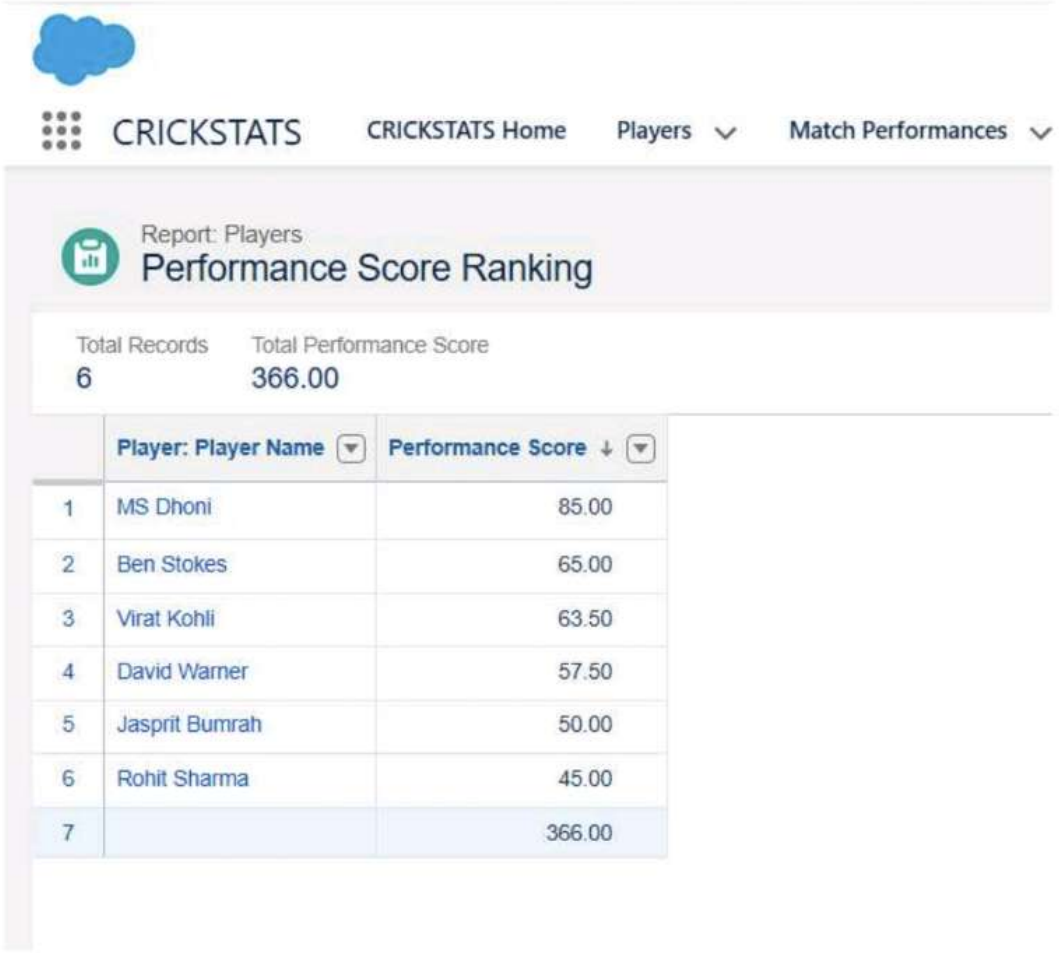
- Top Run Scorers
- Top Wicket Takers
- Performance Score Ranking
- Players by Role

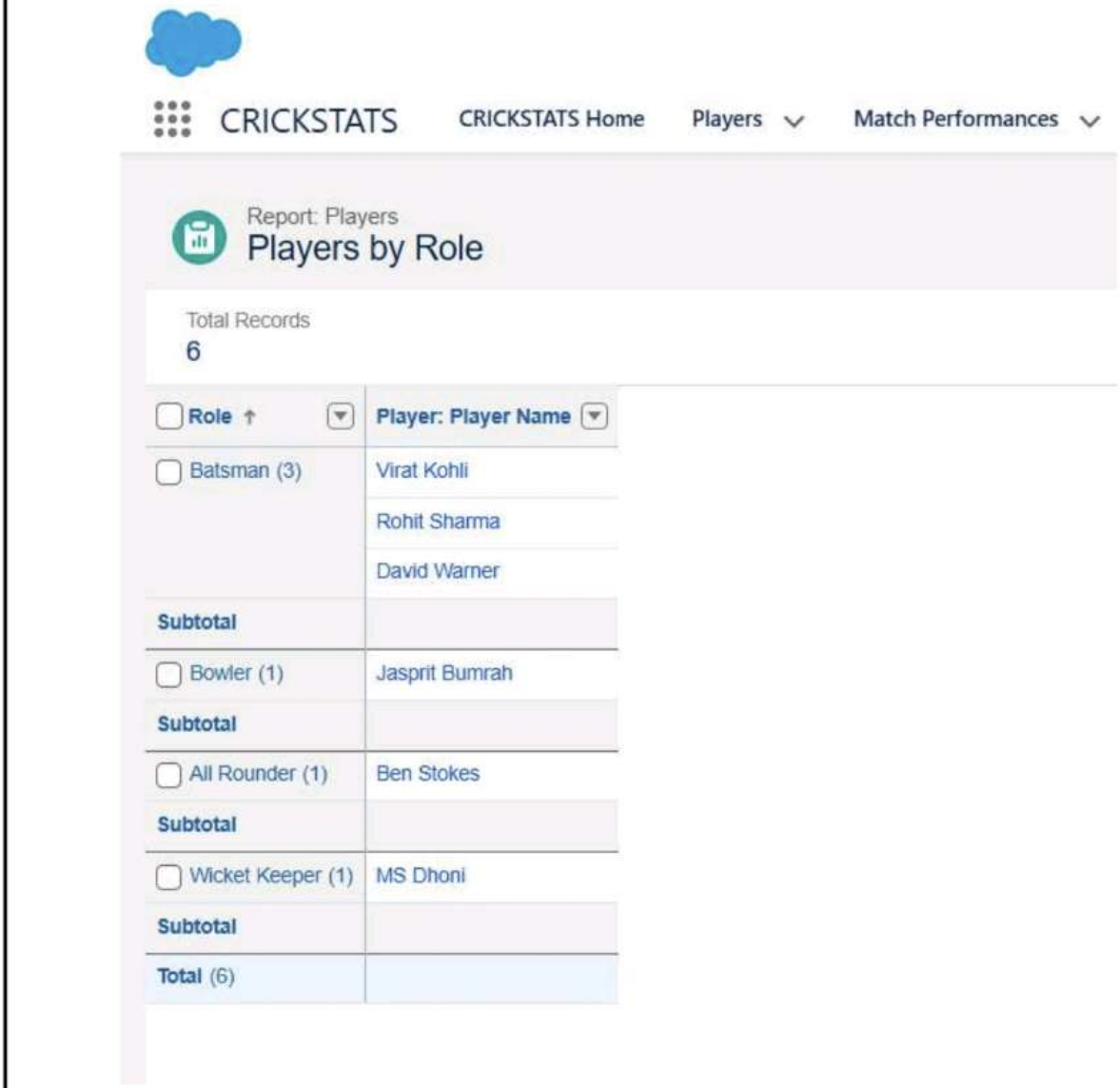
Reports allow quick comparison of player performance.



Player: Player Name	Total Runs	Matches Played	AVG Score	Performance Score
Ben Stokes (1)	70	2	35.00	65.00
Subtotal	70	2	35.00	65.00
David Warner (1)	115	2	57.50	57.50
Subtotal	115	2	57.50	57.50
Jasprit Bumrah (1)	0	2	0.00	50.00
Subtotal	0	2	0.00	50.00
MS Dhoni (1)	110	2	55.00	85.00
Subtotal	110	2	55.00	85.00
Rohit Sharma (1)	90	2	45.00	45.00
Subtotal	90	2	45.00	45.00
Virat Kohli (1)	127	2	63.50	63.50
Subtotal	127	2	63.50	63.50
Total (6)	512	12	256.00	366.00


CRICKSTATS
CRICKSTATS Home
Players
Match Performances
Report: Players
Top Wicket Takers
Total Records
6
Total Total Wickets
11
Player: Player Name
Total Wickets
1 Jasprit Bumrah 5
2 Ben Stokes 3
3 MS Dhoni 3
4 Virat Kohli 0
5 Rohit Sharma 0
6 David Warner 0
7 11


CRICKSTATS
CRICKSTATS Home
Players
Match Performances
Report: Players
Performance Score Ranking
Total Records
6
Total Performance Score
366.00
Player: Player Name
Performance Score
1 MS Dhoni 85.00
2 Ben Stokes 65.00
3 Virat Kohli 63.50
4 David Warner 57.50
5 Jasprit Bumrah 50.00
6 Rohit Sharma 45.00
7 366.00


CRICKSTATS
CRICKSTATS Home
Players
Match Performances
Report: Players
Players by Role
Total Records
6
Role
Player: Player Name
Batsman (3)
Virat Kohli
Rohit Sharma
David Warner
Subtotal
Bowler (1)
Jasprit Bumrah
Subtotal
All Rounder (1)
Ben Stokes
Subtotal
Wicket Keeper (1)
MS Dhoni
Subtotal
Total (6)

DASHBOARD

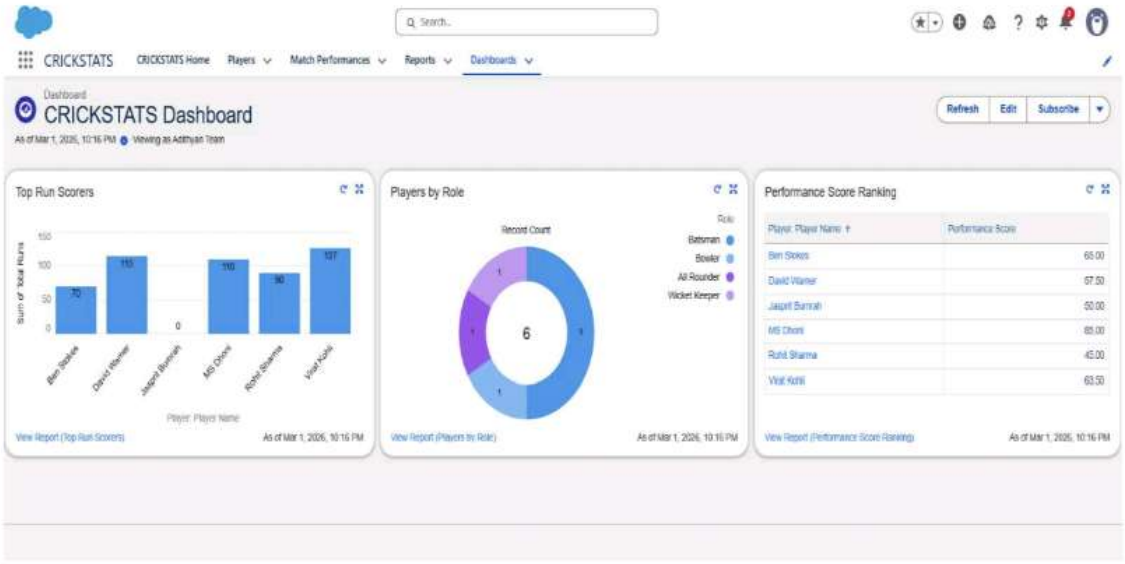
DASHBOARD

A dashboard was created to visually represent performance analytics using charts and tables.

Components include:

- Bar Chart – Top Run Scorers
- Pie Chart – Role Distribution
- Table – Performance Score Ranking

Dashboards provide quick insights into overall player performance.



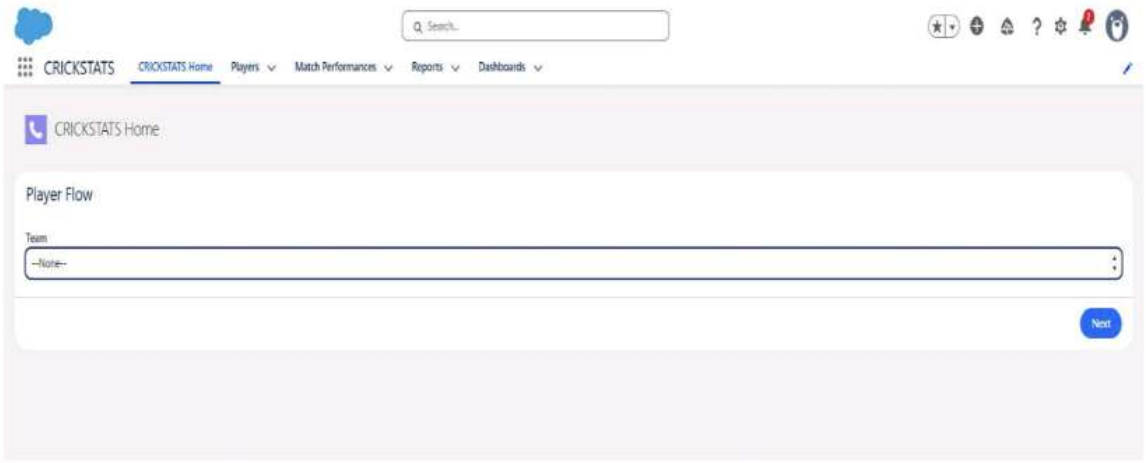
FLOW AUTOMATION

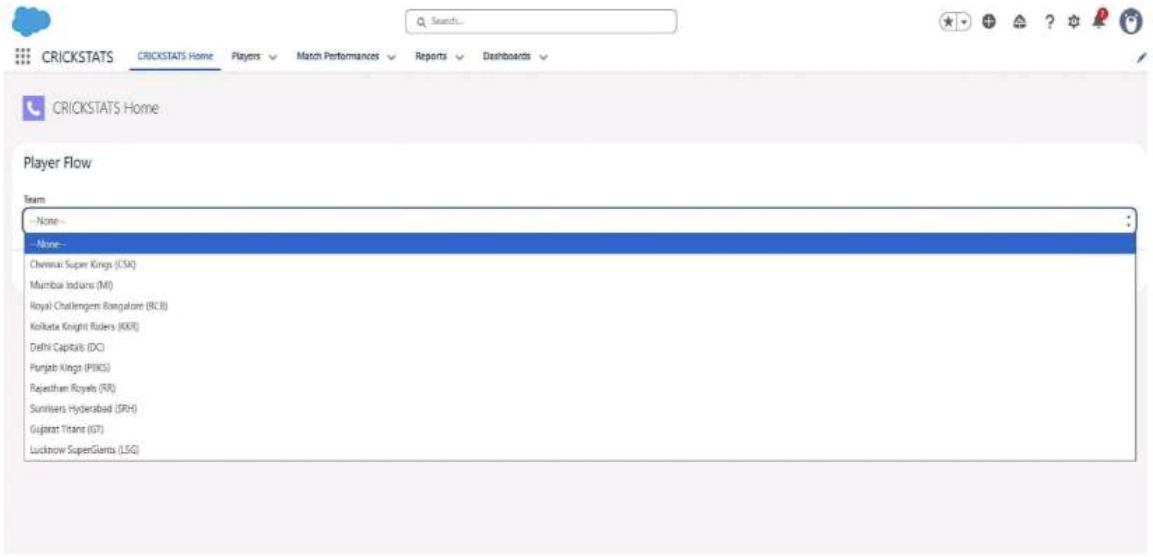
FLOW AUTOMATION

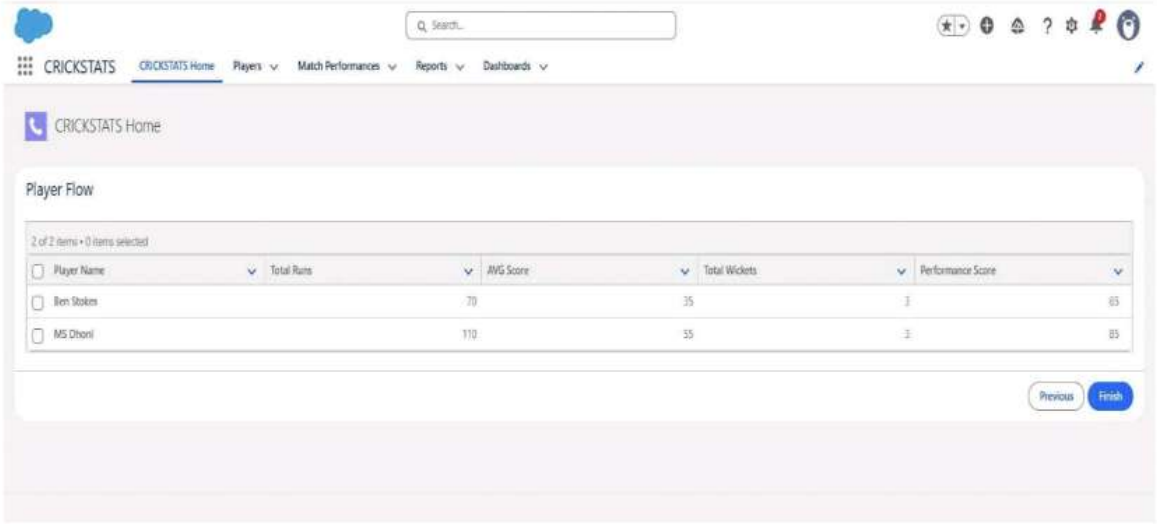
A Screen Flow was implemented to allow users to filter players by team.

The flow performs the following steps:

1. User selects a team from a picklist.
2. The system retrieves players belonging to that team.
3. Filtered results are displayed in a data table.




CRICKSTATS
CRICKSTATS Home
Players
Match Performances
Reports
Dashboards
CRICKSTATS Home
Player Flow
Team
--None--
--None--
Chennai Super Kings (CSK)
Mumbai Indians (MI)
Royal Challengers Bangalore (RCB)
Kolkata Knight Riders (KKR)
Delhi Capitals (DC)
Punjab Kings (PBKS)
Rajasthan Royals (RR)
Sunrisers Hyderabad (SRH)
Gujarat Titans (GT)
Lucknow SuperGiants (LSG)


CRICKSTATS
CRICKSTATS Home
Players
Match Performances
Reports
Dashboards
CRICKSTATS Home
Player Flow
2 of 2 items • 0 items selected
Player Name
Total Runs
AVG Score
Total Wickets
Performance Score
Ben Stokes
70
35
3
65
MS Dhoni
110
55
3
85
Previous
Finish

SYSTEM OUTPUT

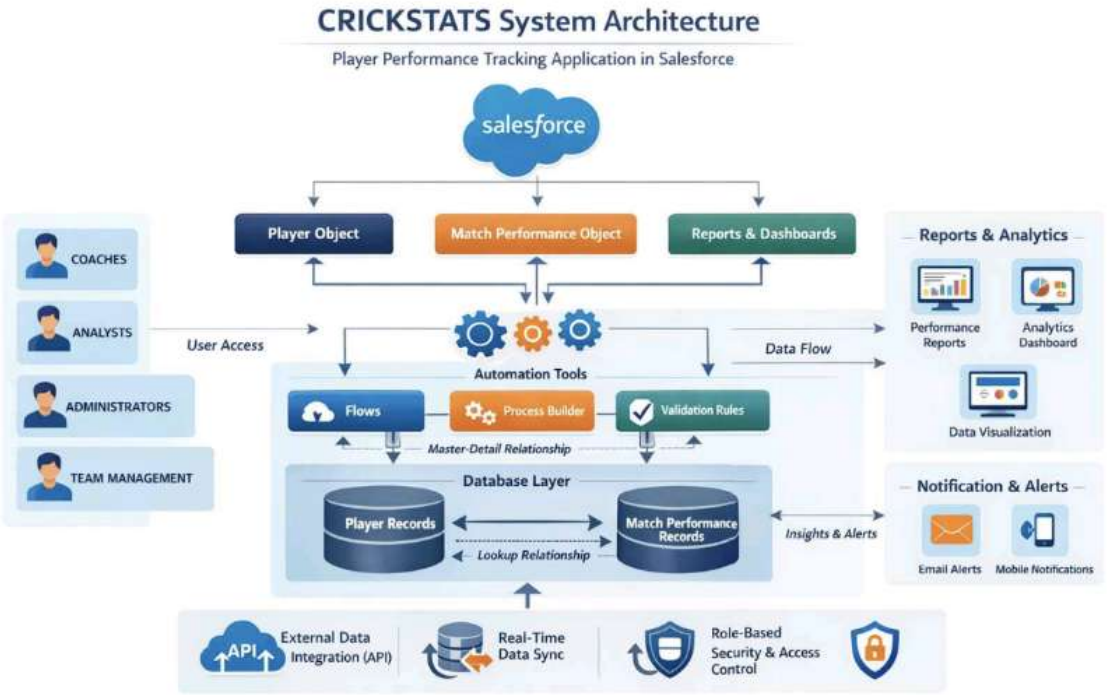
SCREENS

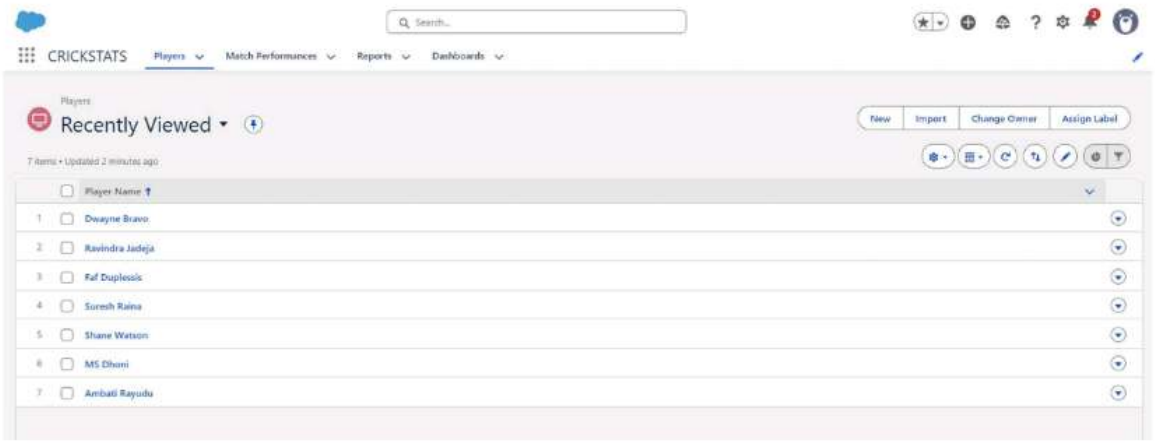
SYSTEM OUTPUT SCREENS

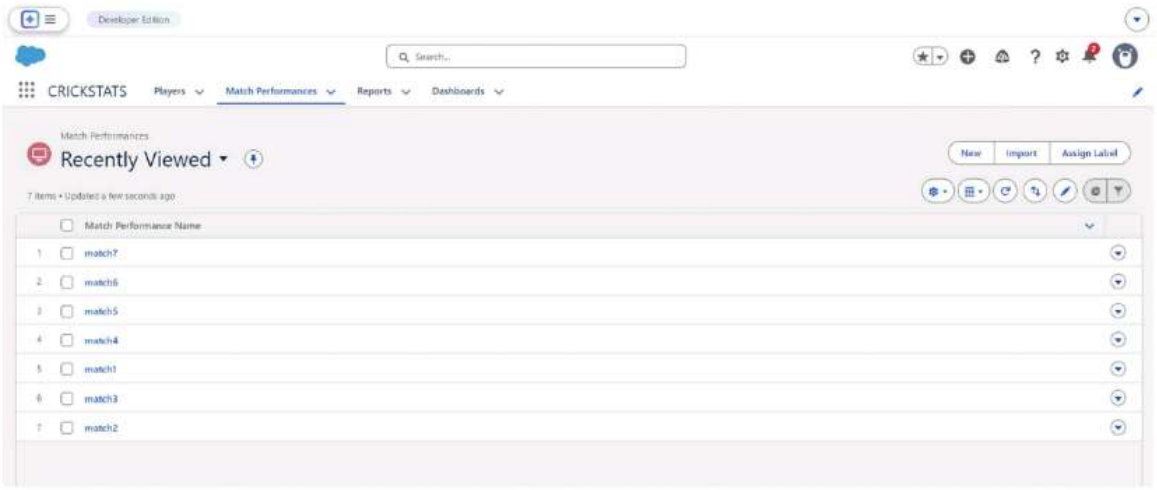
This section displays sample output screens from the system.

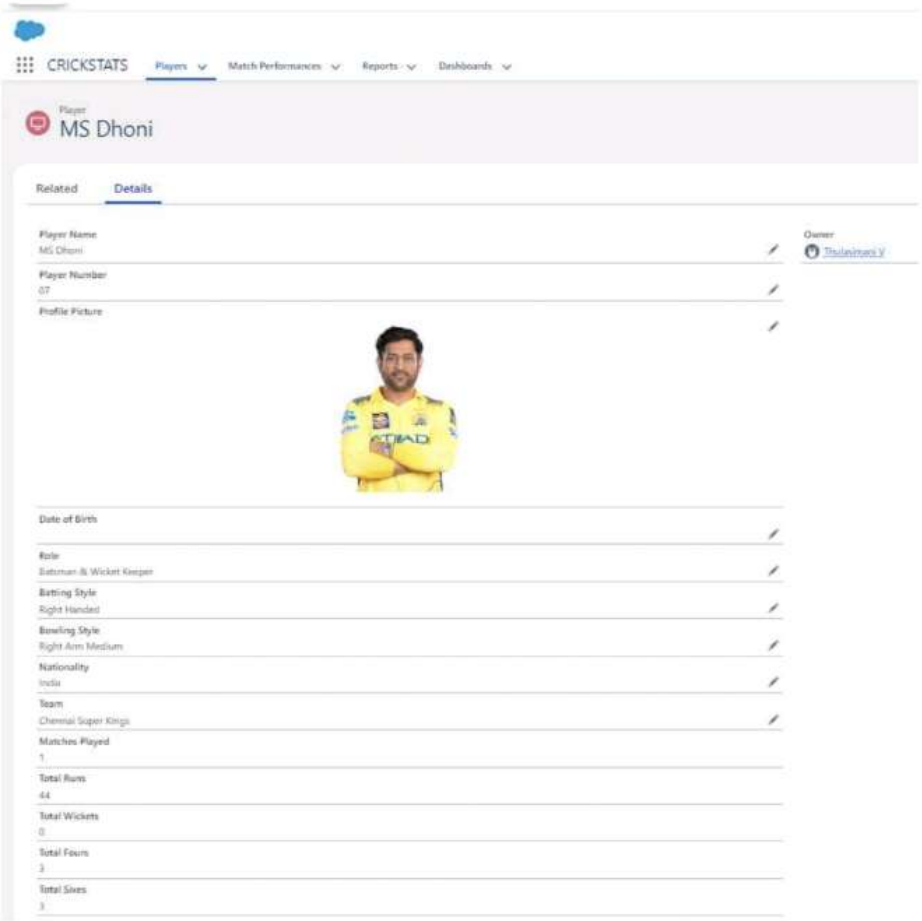
Examples include:

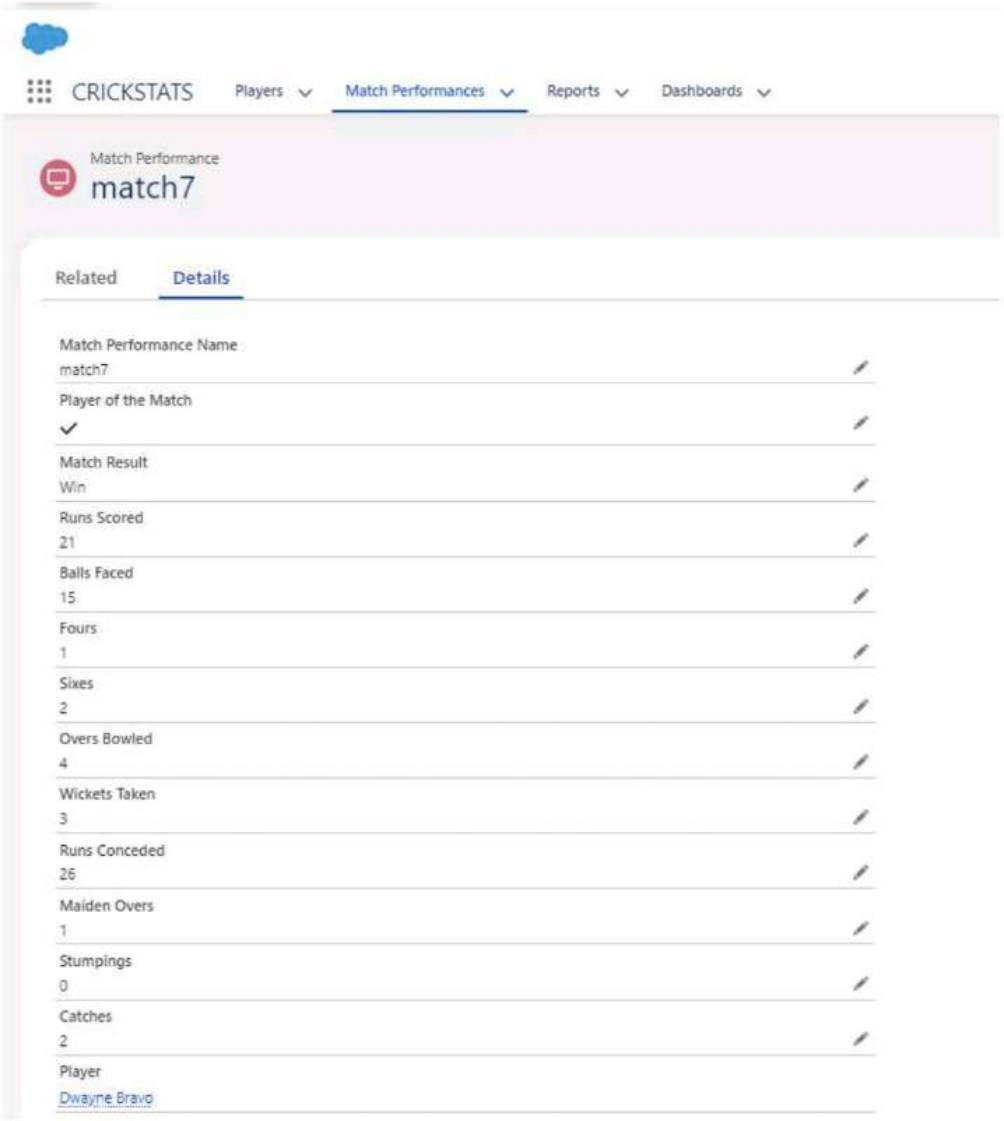
- System Architecture Diagram
- Player Record Page
- Match Performance Record Page
- Player List View
- Match Performance List View
- Dashboard View
- Flow Output




Search...
CRICKSTATS
Players
Match Performances
Reports
Dashboards
Players
Recently Viewed
New
Import
Change Owner
Assign Label
7 items • Updated 2 minutes ago
Player Name
1 Dwayne Bravo
2 Ravindra Jadeja
3 Faf Duplessis
4 Suresh Raina
5 Shane Watson
6 MS Dhoni
7 Ambati Rayudu


Developer Edition
Search...
CRICKSTATS
Players
Match Performances
Reports
Dashboards
Match Performances
Recently Viewed
New
Import
Assign Label
7 items • Updated a few seconds ago
Match Performance Name
1 match7
2 match6
3 match5
4 match4
5 match1
6 match3
7 match2


CRICKSTATS
Players
Match Performances
Reports
Dashboards
Player
MS Dhoni
Related
Details
Player Name
MS Dhoni
Player Number
07
Profile Picture
Date of Birth
Role
Batsman & Wicket Keeper
Batting Style
Right Handed
Bowling Style
Right Arm Medium
Nationality
India
Team
Chennai Super Kings
Matches Played
1
Total Runs
44
Total Wickets
0
Total Fours
3
Total Sixes
3
Owner


CRICKSTATS
Players
Match Performances
Reports
Dashboards
Match Performance
match7
Related
Details
Match Performance Name
match7
Player of the Match
✓
Match Result
Win
Runs Scored
21
Balls Faced
15
Fours
1
Sixes
2
Overs Bowled
4
Wickets Taken
3
Runs Conceded
26
Maiden Overs
1
Stumpings
0
Catches
2
Player
Dwayne Bravo

ENHANCEMENTS IMPLEMENTED

ENHANCEMENTS IMPLEMENTED

Several enhancements were implemented to improve system functionality.

- Automated Strike Rate calculation
- Automated Economy Rate calculation
- Performance Score ranking system
- Advanced roll-up summary fields
- Data validation rules for realistic statistics
- Record type classification
- Custom Lightning App Page
- Interactive filtering through Screen Flow
- Player profile image integration

These enhancements transform the application into a comprehensive cricket performance analytics system.

CONCLUSION

CONCLUSION

The CRICKSTATS platform successfully demonstrates how Salesforce declarative tools can be used to develop a complete performance analytics system. The system automates statistical calculations, ensures data accuracy through validation rules, and provides meaningful insights through reports and dashboards.

By integrating data modelling, automation, and analytics within the Salesforce Lightning environment, the project delivers a scalable and user-friendly solution for cricket performance management.